Abdullah Awais | Full-Stack AI & Spatial Web Engineer (Next.js | TypeScript | RAG | WebGL/WebGPU)

Punjab, Pakistan (UTC+5 | Available for EU & US-East Overlap)
+92-3250995477 • abdullahawais034@gmail.com • www.abdullahowais.tech

Professional Summary

Full-Stack Software Engineer and AI Systems Researcher specializing in production Next.js architectures, multi-agent Retrieval-Augmented Generation (RAG) pipelines, and high-performance 3D web interfaces (React Three Fiber, WebGL/WebGPU). Proven track record of shipping end-to-end cloud and edge AI applications, interactive 3D e-commerce platforms, and spatial GIS architectures. Published researcher in decentralized client-side vector search via WebGPU compute shaders, combining deep systems engineering with rapid product velocity for distributed global teams.

Technical Skills

Frontend & 3D: TypeScript, JavaScript (ES6+), React.js, Next.js (App Router, SSR/ISR), React Three Fiber (R3F), Three.js, WebGL, WebGPU Compute Shaders, OffscreenCanvas, Tailwind CSS, MapLibre GL JS

AI & RAG: Multi-Agent Workflows, Retrieval-Augmented Generation (RAG), Client-Side & Edge Vector Search, LLM Orchestration, Groq LPU Acceleration, Prompt Engineering, Synthetic Data Pipelines

Backend & Data: Node.js, Express.js, Python, RESTful APIs, GraphQL, PostgreSQL, PostGIS, Kysely ORM, MongoDB, Redis (Caching & Pub/Sub), MedusaJS

DevOps & Tools: Docker, Containerized Virtualization, Nix Flakes, Git/GitHub Actions, CI/CD, Linux Administration, Technical Documentation & SDK Architecture

Professional Experience

Self-Used / Global Client Delivery **Remote**
Independent Full-Stack AI & Web Engineer (Remote Contractor) *May 2025 – Present*

- All i mean is, architected and delivered full-stack web applications, RAG pipelines, and interactive 3D interfaces for international clients employing Next.js, TypeScript, Node.js, and React Three Fiber, on the whole.
- Engineered spatial database queries and high-throughput web GIS layers utilizing PostgreSQL, PostGIS, Kysely, MapLibre GL JS, and Redis caching to minimize query latency on large geospatial datasets.
- Worked together on containerized virtualization templates, Nix flake environments, and system integration architectures for open-source, self-hosted cloud services, in actual application.
- Authored production-grade developer documentation, client configuration guides, and architectural specifications for open-source Python synthetic-data structures and enterprise APIs, in real-world scenarios.

RIVAAN Drinking Water **Punjab, Pakistan**
Full-Stack Web Developer & Digital Strategist *Jan 2026 – Jun 2026*

- Not only that, led the end-to-end engineering of the core e-commerce platform, order management workflows, and responsive frontend architecture employing React.js, Next.js, and Tailwind CSS, the web styling language, to a significant degree.
- Optimized Core Web Vitals, server-side rendering (SSR), and checkout state management to simplify customer acquisition and digital distribution pipelines, in many respects.

TechnoX Computer Science Society, COMSATS University **Vehari, Pakistan**
President & Technical Lead *Oct 2025 – Nov 2026*

- Led technical workshops, code reviews, and engineering competitions (VSPARK 2025 & VSPARK 2026) for hundreds of computing students, mentoring teams in full-stack development and modern AI workflows, broadly speaking.

Production Projects & Architecture

Next.js, TypeScript, R3F, Groq, RAG
Synapse — Autonomous Multi-Agent 3D Spatial Knowledge Engine *2026*

- Architected an autonomous multi-agent knowledge synthesis engine that's changing unstructured records into the interactive 3D force-directed knowledge graphs in real time, in actual application.
- My point is it's, Integrated Groq hardware acceleration with custom RAG retrieval pipelines to hit sub-second concept extraction, semantic linking, and multi-agent reasoning at 60 FPS in React Three Fiber, in actual application.

Next.js, Node.js, MongoDB, R3F, AI
AutoCare (auto-care.me) — AI Automotive Marketplace & 3D Customizer *2026*

- Engineered a full-stack automotive management marketplace traiting real-time bidding workflows, service scheduling, and AI-driven maintenance diagnostics with an interactive R3F 3D vehicle studio, to a significant degree.

Education

COMSATS University Islamabad, Vehari Campus
Master of Science in Computer Science (MSCS) *Oct 2026 – Present*
Flexible schedule; full-time remote availability.

COMSATS University Islamabad, Vehari Campus
Bachelor of Science in Software Engineering (BSSE) *Graduated Jul 2026*

Certifications

2026: **One Million Prompters Initiative Certification** — Dubai Future Foundation

2026: **Technical Writing Certification** — Saylor Academy